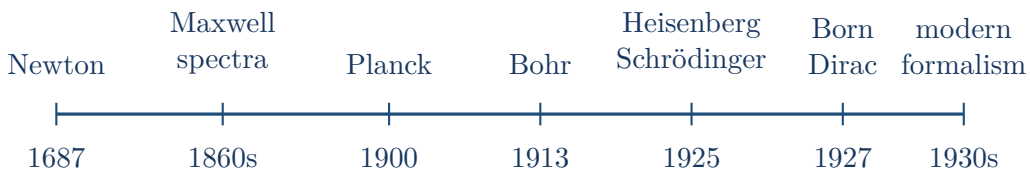
# The Historical Development of Quantum Mechanics

Lecture notes: from classical physics to the quantum revolution




Article-style notes with derivations, context, timelines, and study questions

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# 1 How to read these notes

These notes are not only a chronological list of names and dates. They are organized around a more important question:

**Guiding question**

How did physics arrive at the conclusion that microscopic systems cannot be described by classical trajectories, continuous radiation, and deterministic measurement-independent properties?

Quantum mechanics emerged because several independent research programs converged on the same conclusion. Classical mechanics, classical electromagnetism, thermodynamics, spectroscopy, chemistry, and atomic physics were all successful, but they produced contradictions when applied to microscopic phenomena.

The goal is therefore to understand:

1. what classical physics assumed;
2. which experiments and theoretical problems challenged those assumptions;
3. how the old quantum theory partially solved the problems;
4. why the old quantum theory failed;
5. how the full mathematical structure of quantum mechanics emerged between 1925 and 1927.

**Main historical thesis**

Quantum mechanics was not invented to make physics strange. It was invented because classical physics, when used carefully, predicted results that experiments did not show. The "strangeness" of quantum mechanics is the price of keeping agreement with nature.

# 2 The classical worldview before the quantum crisis

## 2.1 Newtonian mechanics and determinism

The classical image of nature was shaped by Isaac Newton's mechanics. In this view, a physical body has a definite position and velocity at every instant. If the forces are known, the future motion follows from Newton's second law,

$$\mathbf{F} = m\mathbf{a}.$$

A planet, a projectile, a pendulum, or a molecule was expected to move along a definite trajectory.

The deeper philosophical consequence was determinism. If at a given time we know all positions and velocities, and if we know all forces, then the future and the past are determined by differential equations. Pierre-Simon Laplace expressed this ideal in the famous image of an intelligence that, knowing all forces and all positions, could compute the entire future and past of the universe.

## 2.2 Analytical mechanics: Lagrange and Hamilton

In the eighteenth and nineteenth centuries, classical mechanics was reformulated by Lagrange and Hamilton. These formulations did not merely change notation; they introduced concepts

that later became central in quantum theory.

In Hamiltonian mechanics, a system is described by coordinates $q_i$ and conjugate momenta $p_i$. The Hamiltonian $H(q, p, t)$ usually represents the total energy. Hamilton's equations are

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}.$$

The state of a classical system is a point in phase space. Once this point is specified, the later motion is determined.

**Why Hamiltonian mechanics mattered later**

Quantum mechanics did not abandon energy or Hamiltonians. Instead, it transformed them. In quantum mechanics, the Hamiltonian becomes an operator, and the state is no longer a point in phase space but a vector, wavefunction, or density operator.

### 2.3 Classical fields and Maxwell's electromagnetism

The nineteenth century also produced the classical theory of electromagnetic fields. Maxwell's equations unified electricity, magnetism, and optics. In vacuum they imply electromagnetic waves traveling at speed

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}.$$

Because this speed equals the measured speed of light, Maxwell concluded that light is an electromagnetic wave.

This was one of the greatest successes of classical physics. By the end of the nineteenth century, light appeared to be understood as a wave, matter as particles or continuous media, and heat as molecular motion.

### 2.4 The classical assumptions that later failed

The classical worldview contained several assumptions that seemed natural:

1. Energy can vary continuously.
2. Radiation is a continuous electromagnetic wave.
3. Matter consists of objects with definite trajectories.
4. Measurement can in principle reveal pre-existing properties without changing the basic state of the system.
5. Physical properties such as position and momentum can simultaneously have arbitrarily precise values.

Each of these assumptions was challenged by the development of quantum physics.

## 3 The wave-particle problem of light

### 3.1 Newton, Huygens, and early theories of light

From the seventeenth century onward, light was interpreted in two competing ways. Newton favored a corpuscular view: light consists of small particles. Christiaan Huygens proposed a wave theory: light is a disturbance that propagates through space.

For a long time this was a debate between two pictures. The later history of quantum mechanics showed that neither classical picture was sufficient by itself.

### 3.2 Young's double-slit experiment and interference

Thomas Young's double-slit experiment, performed in the early nineteenth century, provided strong evidence for the wave nature of light. Light passing through two narrow openings forms an interference pattern: alternating bright and dark fringes.

For two coherent waves, constructive interference occurs when the path difference is an integer multiple of the wavelength:

$$\Delta L = m\lambda, \qquad m \in \mathbb{Z}.$$

Destructive interference occurs when

$$\Delta L = \left(m + \frac{1}{2}\right)\lambda.$$

Interference is difficult to explain using classical particles moving independently along definite paths. It is natural for waves.

**Classical conclusion around the mid-nineteenth century**

By the time of Maxwell, the wave theory of light seemed decisively established. Light was understood as a continuous electromagnetic wave.

### 3.3 Why this later became a problem

The wave theory explained interference, diffraction, and polarization. However, at the beginning of the twentieth century, experiments such as the photoelectric effect and Compton scattering showed that light also transfers energy and momentum in localized packets.

Thus the history of quantum mechanics begins with a paradox: the best classical theory of light was a wave theory, but some experiments forced light to behave as if it consisted of particles.

## 4 Heat, statistics, and the problem of microscopic matter

### 4.1 Thermodynamics and macroscopic laws

Thermodynamics was developed to understand heat, work, engines, temperature, and entropy. It was extremely successful without requiring detailed knowledge of atoms.

The first law of thermodynamics expresses energy conservation:

$$\Delta U = Q - W,$$

where $U$ is internal energy, $Q$ is heat added to the system, and $W$ is work done by the system.

The second law introduced entropy and the direction of irreversible processes. These laws were macroscopic: they described bulk matter without specifying microscopic motion.

### 4.2 Maxwell, Boltzmann, and statistical mechanics

Maxwell and Boltzmann connected thermodynamics to the molecular theory of matter. Temperature became a measure of average molecular kinetic energy. For a monatomic ideal gas,

$$\langle E_k \rangle = \frac{3}{2} k_{\mathrm{B}} T.$$

This result is one of the triumphs of classical kinetic theory.

Boltzmann also introduced a statistical interpretation of entropy:

$$S = k_{\mathrm{B}} \ln W,$$

where $W$ is the number of microscopic configurations corresponding to a macroscopic state.

### 4.3 The equipartition theorem

Classical statistical mechanics contains the equipartition theorem: each quadratic degree of freedom contributes an average energy of

$$\frac{1}{2} k_{\mathrm{B}} T.$$

For many gases this worked well. But it led to serious failures when applied to radiation and to the heat capacities of solids.

**A hidden crisis**

Classical statistical mechanics was too generous: it allowed energy to be shared continuously among infinitely many modes. This caused wrong predictions for blackbody radiation and for low-temperature heat capacities.

## 5 Spectroscopy and the discovery that atoms have structure

### 5.1 Fraunhofer lines and discrete spectra

In the early nineteenth century, Joseph von Fraunhofer observed dark lines in the solar spectrum. Later, Kirchhoff and Bunsen connected spectral lines with chemical elements. Each element emitted or absorbed light at characteristic wavelengths.

This was one of the earliest signs that atoms possess internal structure. If atoms were simple classical objects, why should they emit only certain sharply defined frequencies?

### 5.2 Balmer's formula

In 1885, Johann Balmer found an empirical formula for visible spectral lines of hydrogen:

$$\lambda = B \frac{n^2}{n^2 - 4}, \qquad n = 3, 4, 5, \ldots$$

Later this was generalized into the Rydberg formula:

$$\frac{1}{\lambda} = R_H \left( \frac{1}{m^2} - \frac{1}{n^2} \right), \qquad n > m.$$

For the Balmer series, $m = 2$.

This formula was mathematically simple and experimentally accurate, but it had no classical explanation at first.

**Spectral lines as a clue**

Discrete spectral lines suggested that atoms could only exchange energy in definite amounts. This was not yet quantum mechanics, but it was one of the strongest clues that atomic energy is not continuous.

### 5.3 Spectroscopy as a bridge between chemistry and physics

Spectroscopy linked atomic physics with chemistry. Chemical elements had unique spectral fingerprints. This supported the idea that atoms were real physical systems with internal degrees of freedom.

The periodic table, spectroscopy, and chemical regularities all pointed to hidden atomic structure. But classical mechanics and electrodynamics did not provide a stable model of the atom.

## 6 Electrons, X-rays, radioactivity, and the breakdown of indivisible atoms

### 6.1 Cathode rays and the electron

In the late nineteenth century, cathode-ray experiments showed that atoms contain smaller charged particles. J. J. Thomson's experiments in 1897 identified the electron and measured its charge-to-mass ratio.

The electron was much lighter than any atom. Therefore atoms were not indivisible. They had internal components.

### 6.2 X-rays

In 1895, Wilhelm Röntgen discovered X-rays. These penetrating rays revealed new forms of radiation and opened a new experimental window into matter. X-rays later became crucial for determining crystal structures and for studying atomic arrangements.

### 6.3 Radioactivity

Henri Becquerel discovered radioactivity in 1896, and Marie Curie and Pierre Curie carried out foundational work on radioactive elements. Radioactivity showed that atoms could transform and emit energetic particles and radiation.

This contradicted the older image of atoms as permanent, indivisible units.

### 6.4 Rutherford scattering and the nuclear atom

Ernest Rutherford and collaborators studied the scattering of alpha particles by thin metal foils. Most alpha particles passed through, but some were deflected through large angles. Rutherford

interpreted this as evidence for a tiny, massive, positively charged nucleus.

The Rutherford model placed electrons around a central nucleus. However, it created an immediate classical problem.

**Classical instability of the Rutherford atom**

According to classical electrodynamics, an accelerating charge radiates energy. An electron orbiting a nucleus is accelerated. It should lose energy, spiral into the nucleus, and the atom should collapse. Stable atoms should not exist.

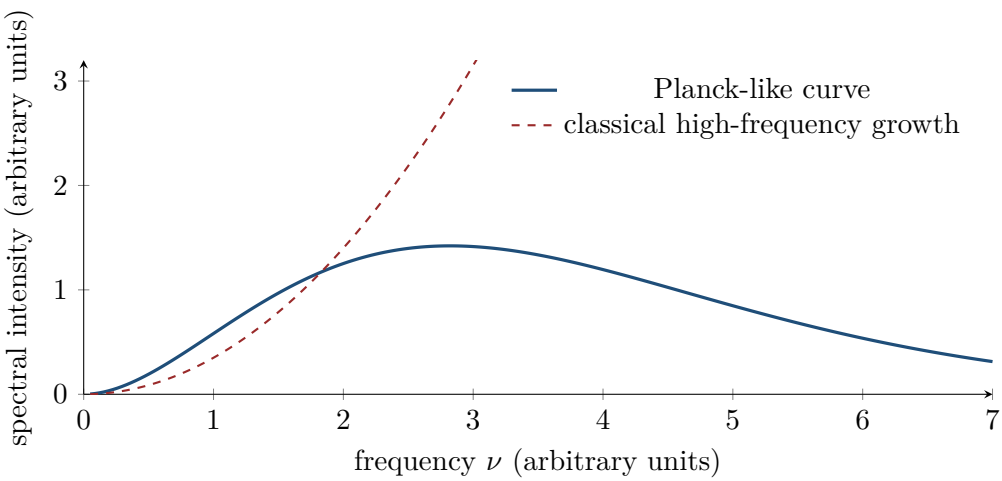
Classical physics could not explain why atoms are stable or why they emit discrete spectral lines.

# 7 Blackbody radiation and Planck's quantum hypothesis

## 7.1 The blackbody problem

A blackbody is an ideal object that absorbs all incident radiation and emits thermal radiation determined only by its temperature. The problem was to find the spectral energy distribution.

Experiments showed that the intensity rises with frequency, reaches a maximum, and then falls at high frequencies. The shape of the spectrum depends on temperature.




## 7.2 Rayleigh-Jeans law and the ultraviolet catastrophe

Using classical electromagnetic modes in a cavity and equipartition of energy, Rayleigh and Jeans obtained a law that worked at low frequencies but failed at high frequencies. In frequency form, the classical result is proportional to

$$u^2 k_{\mathrm{B}} T.$$

As $\nu \to \infty$, the predicted energy density diverges. This is the ultraviolet catastrophe.

**The ultraviolet catastrophe**

Classical physics predicted that a hot object should radiate an infinite amount of energy at high frequencies. Nature does not do this. The failure was not numerical; it was conceptual.

### 7.3 Planck's derivation

In 1900, Max Planck found a formula matching the observed spectrum. To obtain it, he assumed that oscillators exchanging energy with radiation could only have energies

$$E_n = nh\nu, \qquad n = 0, 1, 2, \ldots$$

where $h$ is Planck's constant and $\nu$ is the oscillator frequency.

The average energy of a quantized oscillator is

$$\langle E \rangle = \frac{h\nu}{\exp(h\nu/k_{\mathrm{B}}T) - 1}.$$

Multiplying by the density of electromagnetic modes gives Planck's radiation law.

### 7.4 Why quantization solves the catastrophe

At high frequency, $h\nu$ is large. If $h\nu \gg k_{\mathrm{B}}T$, exciting one quantum of such a mode is thermally unlikely. Therefore high-frequency modes are suppressed by the exponential factor.

Classically, every mode receives energy $k_{\mathrm{B}}T$. Quantum mechanically, high-frequency modes are too expensive to excite.

**Planck's revolutionary step**

Energy exchange is not always continuous. For an oscillator of frequency $\nu$, allowed energies come in units of $h\nu$. This was the first quantum hypothesis.

## 8 Einstein and the light quantum

### 8.1 The photoelectric effect

The photoelectric effect is the emission of electrons from a material when light shines on it. Classical wave theory predicts that the energy delivered to electrons should depend mainly on intensity. Stronger light should eventually eject electrons even at low frequency.

Experiments showed something else:

1. Below a threshold frequency, no electrons are emitted, no matter how intense the light is.
2. Above the threshold frequency, emission occurs almost immediately.
3. The maximum kinetic energy of emitted electrons depends on the frequency, not on the intensity.

### 8.2 Einstein's explanation

In 1905, Albert Einstein proposed that light itself consists of localized energy quanta. Each quantum has energy

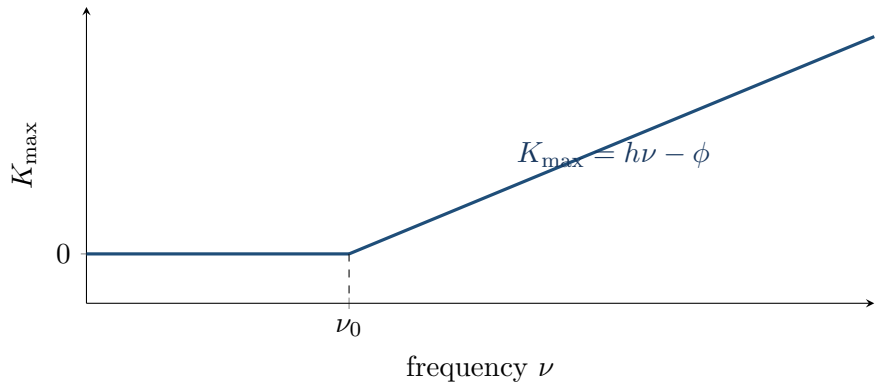
$$E = h\nu.$$

If a light quantum gives its energy to an electron, part of the energy is used to overcome the work function $\phi$, and the rest becomes kinetic energy:

$$K_{\max} = h\nu - \phi.$$

This explains the threshold frequency

$$\nu_0 = \frac{\phi}{h}.$$




### 8.3 Conceptual importance

Einstein went further than Planck. Planck quantized the energy exchange between matter and radiation. Einstein treated light itself as quantized.

This was deeply controversial because light had overwhelming wave evidence. The result was not a return to Newtonian particles. The photon is not a tiny classical bullet. It is a quantum object that displays both wave-like and particle-like features.

## 9 Specific heats and the quantization of matter

### 9.1 Classical prediction for solids

Classically, atoms in a solid can be modeled as oscillators. By equipartition, each atom in a solid should contribute approximately $3k_{\mathrm{B}}$ to the heat capacity, giving the Dulong-Petit law:

$$C_V \approx 3Nk_{\mathrm{B}}.$$

This works well at high temperature for many solids.

But at low temperature, measured heat capacities decrease and approach zero. Classical physics cannot explain this.

### 9.2 Einstein's model of a solid

Einstein applied Planck's quantization to atoms vibrating in a solid. If each atom behaves like a quantized oscillator of frequency $\nu$, its energy levels are separated by $h\nu$.

At low temperature, thermal energy $k_{\mathrm{B}}T$ is too small to excite many quanta. The heat capacity therefore decreases.

### 9.3 Debye's improvement

Peter Debye improved the model by treating a solid as a collection of quantized normal modes with a range of frequencies. Debye's theory explained the low-temperature behavior

$$C_V \propto T^3.$$

**Why specific heats mattered**

Quantization was not only about light. The thermal behavior of solids showed that matter itself has quantized modes of energy.

# 10 Bohr's atom and the old quantum theory

## 10.1 The problem Bohr had to solve

The Rutherford atom had a nucleus and orbiting electrons, but classical electrodynamics made it unstable. Bohr's model of 1913 attempted to explain atomic stability and the hydrogen spectrum.

Bohr introduced postulates that were not derivable from classical physics:

1. Electrons can occupy only certain stationary orbits.
2. In a stationary orbit, the electron does not radiate energy.
3. Radiation is emitted or absorbed only when the electron jumps between stationary states.
4. The frequency of radiation satisfies

$$h\nu = E_i - E_f.$$

## 10.2 Quantization of angular momentum

Bohr assumed that the angular momentum of the electron is quantized:

$$L = mvr = n\hbar, \qquad n = 1, 2, 3, \ldots$$

where

$$\hbar = \frac{h}{2\pi}.$$

For a hydrogen atom, the Coulomb force supplies centripetal acceleration:

$$\frac{mv^2}{r} = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}.$$

Together with $mvr = n\hbar$, this gives allowed radii

$$r_n = a_0 n^2,$$

where $a_0$ is the Bohr radius.

The corresponding energy levels are

$$E_n = -\frac{13.6\,\mathrm{eV}}{n^2}.$$

## 10.3 Explanation of the hydrogen spectrum

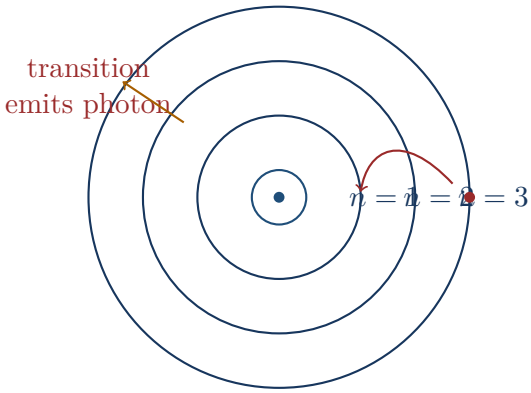
When an electron falls from level $n_i$ to level $n_f$, the emitted photon has energy

$$h\nu = E_{n_i} - E_{n_f}.$$

Using $c = \lambda\nu$, one obtains the Rydberg formula:

$$\frac{1}{\lambda} = R_H \left( \frac{1}{n_f^2} - \frac{1}{n_i^2} \right).$$

This was a major triumph. Bohr's model explained why hydrogen has discrete spectral lines and why the Rydberg formula works.




### 10.4 Successes and limitations

Bohr's model worked impressively for hydrogen-like atoms. It explained spectral regularities and atomic stability in a limited way.

However, it had serious problems:

- It could not correctly describe multi-electron atoms from first principles.
- It mixed classical orbits with non-classical quantum jumps.
- It did not explain transition intensities in a general way.
- It gave no complete theory of measurement.
- It lacked a coherent mathematical foundation.

**The old quantum theory was transitional**

Bohr's model was not modern quantum mechanics. It was a hybrid: classical orbits plus quantum restrictions. Its successes were real, but its foundations were unstable.

## 11 Sommerfeld, quantum numbers, and the extension of the old theory

### 11.1 Elliptical orbits and additional quantum numbers

Arnold Sommerfeld extended Bohr's model by allowing elliptical orbits and introducing additional quantum numbers. The general quantization condition was written as

$$\oint p_i \, \mathrm{d}q_i = n_i h.$$

This connected the old quantum theory with Hamiltonian mechanics.

Sommerfeld's extension helped explain fine structure in hydrogen spectra, especially when relativistic corrections were included.

### 11.2 The correspondence principle

Bohr introduced the correspondence principle: quantum theory should reproduce classical physics in the limit of large quantum numbers.

This principle was historically important because physicists did not yet have the full quantum formalism. It served as a guide for guessing quantum rules from classical formulas.

### 11.3 The limits of quantized orbits

The old quantum theory was increasingly complicated. It required special rules for each problem. It worked best for periodic systems, but failed for many atoms and for non-periodic motion.

A deeper theory was needed: one that did not begin with classical trajectories and then restrict them artificially.

## 12 Experimental pressure: Franck-Hertz, Stern-Gerlach, Compton

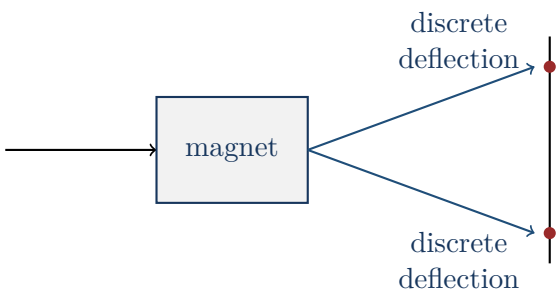
### 12.1 The Franck-Hertz experiment

The Franck-Hertz experiment provided direct evidence that atoms absorb energy in discrete amounts. Electrons passing through mercury vapor lost energy only when their kinetic energy reached certain thresholds. This supported the idea of discrete atomic energy levels.

### 12.2 The Stern-Gerlach experiment

In 1922, Otto Stern and Walther Gerlach sent silver atoms through an inhomogeneous magnetic field. Classically, one might expect a continuous distribution of deflections corresponding to different orientations of magnetic moments.

Instead, the beam split into distinct components.




This experiment later became central for understanding spin and quantization of angular momentum.

### 12.3 Compton scattering

Arthur Compton observed that X-rays scattered by electrons change wavelength. The shift is described by

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta).$$

This formula follows naturally if light quanta carry momentum

$$p = \frac{h}{\lambda}.$$

Compton scattering strengthened the photon concept. Light has not only quantized energy but also particle-like momentum exchange.

## 13 Matter waves and de Broglie's hypothesis

### 13.1 Symmetry between light and matter

If light, traditionally considered a wave, can behave like particles, Louis de Broglie asked whether matter, traditionally considered particles, can behave like waves.

In 1924 he proposed that a particle with momentum $p$ has wavelength

$$\lambda = \frac{h}{p}.$$

This is the de Broglie wavelength.

### 13.2 Connection with Bohr's quantization

Bohr's angular momentum condition can be interpreted as a standing wave condition. For a circular orbit, the circumference must contain an integer number of wavelengths:

$$2\pi r = n\lambda.$$

Using $\lambda = h/p = h/(mv)$, we obtain

$$2\pi r = n\frac{h}{mv} \quad \Rightarrow \quad mvr = n\hbar.$$

Thus Bohr's quantization condition can be understood as a wave condition.

**Matter waves**

The electron is not simply a tiny planet orbiting the nucleus. Its allowed states are connected with wave-like structure. Stable atomic states resemble standing waves more than classical orbits.

### 13.3 Electron diffraction

The wave nature of matter was confirmed experimentally by electron diffraction. Electrons scattered from crystals produce diffraction patterns, just as waves do.

This result made it impossible to keep a purely particle-like description of matter.

# 14 Matrix mechanics: Heisenberg, Born, and Jordan

## 14.1 Heisenberg's problem

By the mid-1920s, the old quantum theory was failing. Werner Heisenberg proposed a radical strategy: do not try to describe unobservable electron orbits. Instead, build the theory from observable quantities such as spectral frequencies and transition amplitudes.

This was a major conceptual shift. The theory should not assume hidden classical trajectories if those trajectories cannot be observed.

## 14.2 Non-commuting quantities

In Heisenberg's approach, physical quantities were represented by arrays of transition quantities. Max Born recognized that these arrays were matrices.

Matrix multiplication is generally non-commutative:

$$AB \neq BA.$$

For position and momentum, the fundamental commutation relation became

$$\hat{x}\hat{p} - \hat{p}\hat{x} = \mathrm{i}\hbar.$$

More compactly,

$$[\hat{x}, \hat{p}] = \mathrm{i}\hbar.$$

**A new mathematical structure**

Classical quantities became non-commuting operators. This was not a small correction to classical mechanics; it was a new kinematic framework.

## 14.3 Energy levels from operators

In matrix mechanics, the allowed energies of a system are obtained from an eigenvalue problem. For the Hamiltonian operator $\hat{H}$, stationary states satisfy

$$\hat{H}|\psi\rangle = E|\psi\rangle.$$

This is the abstract form of a principle that appears throughout quantum mechanics: measurable values correspond to eigenvalues of operators.

# 15 Wave mechanics: Schrödinger's equation

## 15.1 Schrödinger and matter waves

Inspired by de Broglie's matter waves, Erwin Schrödinger developed wave mechanics in 1926. Instead of matrices, he used wavefunctions.

For a nonrelativistic particle of mass $m$ in a potential $V(\mathbf{r}, t)$, the time-dependent Schrödinger equation is

$$\mathrm{i}\hbar\frac{\partial\psi}{\partial t} = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}, t)\right]\psi.$$

For time-independent potentials, stationary states satisfy

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r})\right]\psi = E\psi.$$

### 15.2 Why the equation explains quantization

Quantization appears naturally in boundary-value problems. For example, a particle in a one-dimensional box of length $L$ has wavefunctions satisfying

$$\psi(0) = 0, \qquad \psi(L) = 0.$$

Only certain wavelengths fit into the box:

$$\lambda_n = \frac{2L}{n}.$$

Using $p = h/\lambda$, the allowed energies are

$$E_n = \frac{n^2h^2}{8mL^2}, \qquad n = 1, 2, 3, \ldots$$

Discrete energies are not imposed artificially; they result from wave boundary conditions.

### 15.3 Equivalence with matrix mechanics

Schrödinger's wave mechanics and Heisenberg's matrix mechanics looked very different at first. Soon it was shown that they are mathematically equivalent formulations of the same theory.

This was a decisive moment. Quantum mechanics was no longer a collection of tricks. It had a coherent mathematical structure.

## 16 Born's probability interpretation

### 16.1 What is the wavefunction?

Schrödinger initially hoped that the wavefunction might describe a real spread-out charge distribution. But this interpretation ran into difficulties.

Max Born proposed the probability interpretation: the wavefunction gives probability amplitudes. The probability density for finding a particle near position $\mathbf{r}$ is

$$|\psi(\mathbf{r}, t)|^2.$$

The normalization condition is

$$\int |\psi(\mathbf{r}, t)|^2 \, \mathrm{d}^3r = 1.$$

**Probability enters fundamentally**

In quantum mechanics, probability is not merely ignorance of hidden classical details in the same way as in classical statistical mechanics. The theory itself predicts probabilities for measurement outcomes.

### 16.2 Amplitudes, not ordinary probabilities

Quantum theory does not add ordinary probabilities directly. It adds complex amplitudes. Interference arises because amplitudes can have phases and can cancel.

For two alternatives with amplitudes $A_1$ and $A_2$, the probability is

$$P = |A_1 + A_2|^2,$$

not generally

$$|A_1|^2 + |A_2|^2.$$

This is the mathematical root of quantum interference.

## 17 Uncertainty, complementarity, and the measurement problem

### 17.1 Heisenberg's uncertainty principle

The non-commutation of position and momentum leads to the uncertainty relation

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}.$$

This does not simply mean that instruments are imperfect. It means that quantum states cannot in general assign arbitrarily sharp values to both position and momentum.

### 17.2 Complementarity

Niels Bohr introduced the idea of complementarity. Certain experimental arrangements reveal wave-like behavior; others reveal particle-like behavior. These descriptions are mutually exclusive in a single experimental setup but jointly necessary for a complete account of quantum phenomena.

### 17.3 Measurement and collapse

The Schrödinger equation gives continuous, deterministic evolution of the wavefunction. But measurement gives definite outcomes with probabilities. The transition from a superposition to a definite outcome became known as collapse in the Copenhagen-style interpretation.

This raised deep questions:

- What counts as a measurement?
- Is collapse a physical process or an update of information?
- Are quantum probabilities fundamental or incomplete?

These questions did not stop quantum mechanics from being enormously successful, but they remain philosophically important.

## 18 Spin, Pauli's principle, and the structure of matter

### 18.1 The need for a new quantum number

Spectra showed features that could not be explained by orbital motion alone. The Stern-Gerlach experiment and atomic fine structure suggested an intrinsic angular momentum.

Electron spin was introduced as a new quantum property. It is not literally a classical spinning ball. It is intrinsic angular momentum with no classical analogue.

For an electron,

$$s = \frac{1}{2},$$

and measurement of spin along a chosen axis gives two possible values:

$$S_z = \pm\frac{\hbar}{2}.$$

### 18.2 Pauli exclusion principle

Wolfgang Pauli formulated the exclusion principle: no two identical fermions can occupy the same quantum state.

For electrons in atoms, this principle explains the shell structure of atoms and the organization of the periodic table. Chemistry became understandable from quantum principles.

**Quantum mechanics and chemistry**

The periodic table is not an arbitrary list. Its structure follows from quantum states, electron spin, and the Pauli exclusion principle.

## 19 Quantum statistics: Bose, Einstein, Fermi, and Dirac

### 19.1 Indistinguishability

In classical physics, identical particles can still be imagined as individually labeled. In quantum mechanics, identical particles are fundamentally indistinguishable. This changes statistical counting.

### 19.2 Bosons and Bose-Einstein statistics

Satyendra Nath Bose derived Planck's radiation law using a new counting method for photons. Einstein extended the idea to material particles. Particles with integer spin are bosons and can occupy the same quantum state in large numbers.

This leads to Bose-Einstein statistics and, at sufficiently low temperatures, Bose-Einstein condensation.

### 19.3 Fermions and Fermi-Dirac statistics

Particles with half-integer spin are fermions. They obey the Pauli exclusion principle and Fermi-Dirac statistics.

Electrons, protons, and neutrons are fermions. Their statistics explain electron degeneracy pressure, the structure of metals, and the stability of matter.

# 20 Dirac, relativity, and the modern formalism

## 20.1 Dirac's synthesis

Paul Dirac unified important strands of quantum theory. His transformation theory clarified the relation between matrix mechanics and wave mechanics. He introduced a powerful abstract notation and developed a relativistic equation for the electron.

The Dirac equation predicted the existence of antimatter. The positron was later observed experimentally.

## 20.2 Hilbert space and operators

The modern formalism describes states as vectors in a Hilbert space. Observables are represented by operators. If $\hat{A}$ is an observable, possible measurement outcomes are eigenvalues $a$ satisfying

$$\hat{A}|a\rangle = a|a\rangle.$$

The probability of obtaining outcome $a$ in state $|\psi\rangle$ is related to the projection amplitude

$$|\langle a|\psi\rangle|^2.$$

## 20.3 The Hamiltonian as generator of time evolution

The Schrödinger equation can be written abstractly as

$$\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle.$$

Thus the Hamiltonian remains central, but its role is quantum: it generates time evolution of state vectors.

# 21 What exactly changed from classical to quantum physics?

## 21.1 From trajectories to states

Classical mechanics describes particles by trajectories $\mathbf{r}(t)$. Quantum mechanics describes systems by states, such as wavefunctions $\psi$ or vectors $|\psi\rangle$. A quantum state gives probabilities for measurement outcomes, not a pre-existing list of all classical properties.

## 21.2 From continuous energy to spectra

Many quantum systems have discrete energy spectra. This is especially clear for bound systems such as atoms. Quantization arises naturally from the mathematical structure of states and boundary conditions.

## 21.3 From commuting variables to operators

Classically, quantities such as $x$ and $p$ are numbers. Multiplication commutes. Quantum mechanically, observables are operators, and operator order matters:

$$[\hat{x}, \hat{p}] = \mathrm{i}\hbar.$$

This is the algebraic basis of uncertainty relations.

### 21.4 From ordinary probabilities to probability amplitudes

Quantum theory uses complex amplitudes. Probabilities are squared moduli of amplitudes. This explains interference and makes quantum probability structurally different from classical ignorance.

## 22 A chronological map of the development

| Period | Main development | Why it mattered |
|---|---|---|
| 1687 | Newtonian mechanics | Established deterministic laws of motion and the ideal of trajectories. |
| 18th–19th c. | Lagrange and Hamilton | Introduced analytical mechanics, action, Hamiltonians, and phase space. |
| 1801 | Young's interference | Supported wave theory of light. |
| 1860s | Maxwell's electromagnetism | Identified light as an electromagnetic wave. |
| Mid–late 19th c. | Statistical mechanics | Connected heat with microscopic motion but later produced failures. |
| 1814–1880s | Spectral lines, Balmer, Rydberg | Revealed discrete atomic regularities needing explanation. |
| 1895–1897 | X-rays, radioactivity, electron | Showed atoms and radiation had unexpected internal structure. |
| 1900 | Planck's quantum hypothesis | Introduced discrete energy exchange $E = h\nu$. |
| 1905 | Einstein's light quantum | Explained the photoelectric effect using photons. |
| 1911 | Rutherford atom | Established the nuclear atom but created an instability problem. |
| 1913 | Bohr atom | Explained hydrogen spectra using stationary states and quantum jumps. |
| 1916–1924 | Sommerfeld and old quantum theory | Extended quantization but exposed its limits. |
| 1922–1923 | Stern-Gerlach and Compton | Strengthened evidence for discrete angular momentum and photon momentum. |
| 1924 | de Broglie matter waves | Proposed $\lambda = h/p$, linking particles with waves. |
| 1925 | Heisenberg matrix mechanics | Replaced orbits with observable transition quantities and non-commuting operators. |
| 1926 | Schrödinger wave mechanics | Gave a wave equation for matter and explained quantization through boundary conditions. |
| 1926 | Born interpretation | Interpreted $\|\psi\|^2$ as probability density. |
| 1927 | Uncertainty and complementarity | Clarified limits of classical concepts. |
| 1928 onward | Dirac and modern formalism | Unified quantum mechanics with relativity and abstract operator methods. |

# 23 Conceptual chain: how the pieces led to quantum mechanics

The development can be compressed into the following chain of necessity:

1. Classical mechanics gave deterministic trajectories.
2. Maxwell's theory made light a wave.
3. Thermodynamics and statistical mechanics connected heat to microscopic motion.
4. Spectroscopy showed that atoms emit and absorb only selected frequencies.
5. Blackbody radiation showed that classical continuous energy exchange fails.
6. Planck introduced energy quanta.
7. Einstein treated light as composed of quanta.
8. Rutherford's nuclear atom required a new explanation of stability.
9. Bohr introduced stationary states and quantum jumps.
10. Experiments confirmed discrete atomic energy exchange and photon momentum.
11. de Broglie extended wave-particle duality to matter.
12. Heisenberg removed unobservable orbits and introduced non-commuting quantities.
13. Schrödinger gave a wave equation for quantum states.
14. Born interpreted the wavefunction probabilistically.
15. The modern theory replaced classical trajectories with state vectors, operators, amplitudes, and measurement probabilities.

**The shortest correct summary**

Quantum mechanics developed because atoms, radiation, and microscopic matter displayed stable discrete structures, interference, and energy exchange rules that could not be reconciled with classical continuous trajectories and fields.

# 24 Common misconceptions

## 24.1 Misconception 1: quantum mechanics began with Schrödinger

Schrödinger's equation was a central breakthrough, but it came after decades of problems: blackbody radiation, spectra, the photoelectric effect, atomic instability, and matter waves.

## 24.2 Misconception 2: Planck immediately understood the meaning of quanta

Planck introduced quantization to solve blackbody radiation, but the deeper interpretation developed gradually. Einstein's light quantum, Bohr's atom, and later quantum mechanics changed the conceptual meaning of Planck's idea.

## 24.3 Misconception 3: Bohr's model is modern quantum mechanics

Bohr's model is historically important but conceptually intermediate. It still uses classical orbits and adds quantum rules by hand. Modern quantum mechanics does not require electron orbits in atoms.

### 24.4 Misconception 4: uncertainty is only about measurement disturbance

Measurement disturbance can be relevant, but the uncertainty principle is deeper. It follows from the structure of quantum states and the non-commutation of operators.

### 24.5 Misconception 5: wave-particle duality means objects switch between being waves and particles

A quantum object is not sometimes a classical wave and sometimes a classical particle. Rather, wave and particle are classical concepts that describe different experimental aspects of a more general quantum object.

## 25 Study guide

### 25.1 Questions for review

1. What assumptions of classical mechanics are challenged by quantum mechanics?
2. Why was blackbody radiation a crisis for classical physics?
3. How does Planck's quantization suppress high-frequency radiation?
4. Why does the photoelectric effect support the photon idea?
5. What problem did Bohr's model solve, and what problems did it leave unsolved?
6. How does de Broglie's relation reinterpret Bohr's angular momentum quantization?
7. Why was Heisenberg's rejection of unobservable orbits historically important?
8. What is the meaning of the commutator $[\hat{x}, \hat{p}] = \mathrm{i}\hbar$?
9. Why is $|\psi|^2$ interpreted as a probability density?
10. What is the difference between the old quantum theory and modern quantum mechanics?

### 25.2 Essential equations to remember

| | |
|---|---|
| $E = h\nu$ | Planck-Einstein relation, |
| $K_{\max} = h\nu - \phi$ | photoelectric effect, |
| $\lambda = \dfrac{h}{p}$ | de Broglie wavelength, |
| $L = n\hbar$ | Bohr angular momentum condition, |
| $E_n = -\dfrac{13.6\,\mathrm{eV}}{n^2}$ | hydrogen energy levels, |
| $[\hat{x}, \hat{p}] = \mathrm{i}\hbar$ | canonical commutation relation, |
| $\Delta x\,\Delta p \geq \dfrac{\hbar}{2}$ | uncertainty principle, |
| $\mathrm{i}\hbar\dfrac{\partial\psi}{\partial t} = \hat{H}\psi$ | Schrödinger equation, |
| $P(\mathbf{r}, t) = \lvert\psi(\mathbf{r}, t)\rvert^2$ | Born probability rule. |

## 26 Concluding synthesis

The road to quantum mechanics was long because the classical worldview was powerful and successful. Newtonian mechanics, Maxwell's electromagnetism, and thermodynamics were not naive theories. They described enormous domains of experience with great precision.

The quantum revolution happened because the microscopic world repeatedly refused to fit the classical categories. Radiation was wave-like yet exchanged energy as quanta. Matter was particle-like yet diffracted like waves. Atoms were stable despite classical electrodynamics predicting collapse. Spectra were discrete, not continuous. Heat capacities and blackbody radiation contradicted equipartition.

The final theory that emerged in the 1920s replaced the classical idea of a fully determined trajectory with a new structure: states in Hilbert space, observables as operators, probabilities from amplitudes, and dynamics generated by the Hamiltonian. It preserved classical physics as an approximation in appropriate limits, but it changed the foundations of physical description.

**The lesson of the history**

Quantum mechanics was not a rejection of rational physics. It was the continuation of rational physics under the pressure of experiment. The theory became strange because nature forced physicists to give up classical assumptions that had silently been treated as obvious.

## Suggested further reading

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